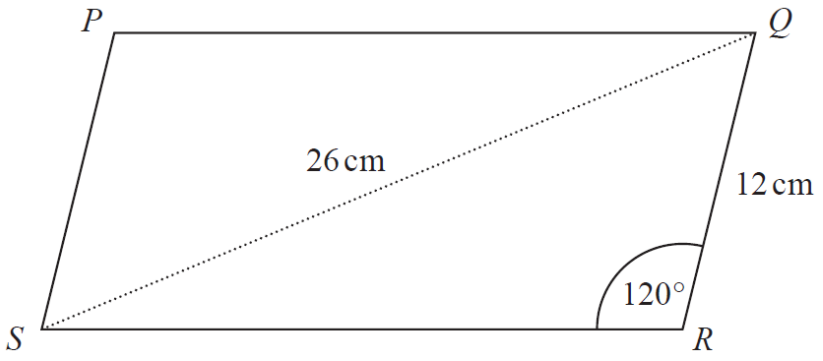
17



Calculate the area of the parallelogram *PQRS*.
Give your answer correct to 3 significant figures.
You must show all your working.

In triangle QRS, use the sine rule:

1 mark
Use sine rule.

$$\sin QSR = \frac{12 \sin 120°}{26}$$

$QSR = \sin^{-1}(0.399704\ldots) = 23.559677\ldots°$

1 mark
Find angle QSR.

$SQR = 180° - 120° - 23.559677\ldots°$

$= 36.440322\ldots°$

1 mark
Triangle angle sum.

Area triangle QRS = ½ × 26 × 12 × sin 36.440322…°

$= 92.66169011\ldots \text{ cm}^2$

1 mark
½ab sin C.

Parallelogram area = 2 × 92.66169011…

$= 185.3233802\ldots \approx 185 \text{ cm}^2$ (3 s.f.)

1 mark
Double; 185 cm².

..... cm²

(Total for Question 17 is 5 marks)